# Sahaj Bhadja

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## Summary

Software Engineer with 2+ years building and supporting **C#/.NET applications, SQL-backed services, distributed integrations, and production automation** with strong debugging, observability, incident ownership, and customer communication.

## Education

**University of Southern California, Los Angeles, CA** **August 2024 – May 2026**
*Master of Science in Computer Science* *GPA: 3.75/4*
Coursework: Database Systems, Web Technologies, Operating Systems, Analysis of Algorithms

**Pandit Deendayal Energy University, Gandhinagar, India** **November 2020 – May 2024**
*Bachelor of Technology in Computer Science and Engineering* *GPA: 3.88/4*
Coursework: Software Engineering, Data Structures, Object-Oriented Programming, Computer Networks

## Experience

**USC Dornsife** **September 2025 – Present**
*Software Engineer* *Los Angeles, CA*

- Reworked asynchronous FastAPI orchestration with explicit job states and failure isolation, processing 1K+ concurrent AI jobs at 3x throughput while sustaining **production reliability, monitoring, incident diagnosis, and service supportability**.
- Optimized indexed Azure SQL queries and row-level isolation for 500+ users, cutting latency from 1.2 seconds to 350 milliseconds through **SQL troubleshooting, data analysis, root-cause investigation, and permanent performance fixes**.

**Grad8** **June 2025 – August 2025**
*Software Engineer Intern* *Los Angeles, CA*

- Delivered React and Flask workflows for uploads, grading history, model controls, and feedback, supporting 150+ instructors and 10K+ daily requests through **API debugging, application tracing, user communication, and production releases**.
- Built concurrent Python ingestion for PDF, DOCX, ZIP, OCR, and multimodal routing, processing 1K+ files per batch while reducing turnaround 60% through **failure analysis, automated testing, and workflow validation**.

**Aasma Technology Solutions** **August 2023 – July 2024**
*Software Developer Engineer - (SecureSync)* *India*

- Designed object-oriented C#/.NET services and large-scale API integrations, improving response times 30% through caching, modular components, and **distributed request tracing while reducing recurring technical debt 20% across production workflows**.
- Automated 20+ unit and integration suites across Windows, Linux, and macOS using Mockito, PyTest, Jenkins, and virtual machines, strengthening **issue reproduction, code-fix validation, CI/CD, and cross-team incident resolution**.

**Aasma Technology Solutions** **May 2023 – August 2023**
*Full Stack Developer Intern - (VibrantChem)* *India*

- Built eight C# and ASP.NET Core microservices with Entity Framework and Azure SQL for authentication, billing, permissions, and notifications, synchronizing 2M+ records through **secure APIs and role-based authorization**.
- Developed React and TypeScript interfaces with WebSocket updates while automating Docker and Jenkins delivery, reducing deployment time 50% through **Sentry monitoring, structured debugging, and reliable multi-environment releases**.

**LST Capitals** **November 2022 – January 2023**
*Web Developer Intern* *India*

- Translated customer requirements into technical specifications for 12+ integrations, mapping booking, payment, coupon, and status workflows to **trace application behavior, clarify system boundaries, and reduce design errors 25%**.
- Built Flask payment APIs supporting 50K+ records with sub-200-millisecond responses while providing **production troubleshooting, customer updates, integration testing, and resolution follow-up** across client demonstrations and operational reviews.

## Skills

**Languages & Frameworks:** C#, .NET, ASP.NET Core, Entity Framework, Python, TypeScript, JavaScript, React, SQL, PowerShell
**Production Support:** Incident Response, Root-Cause Analysis, On-Call Support, Debugging, Log Analysis, Application Tracing, Troubleshooting
**Integrations & Security:** REST APIs, JSON, Webhooks, SSO, OAuth2, JWT, Authentication, Authorization, RBAC, Service Integrations
**Data & Monitoring:** SQL Server, Azure SQL, PostgreSQL, MySQL, Splunk, New Relic, LogicMonitor, Sentry, Kibana, Azure Monitor
**Engineering:** Distributed Systems, Microservices, Git, Docker, Jenkins, CI/CD, Unit Testing, Integration Testing, Agile, Customer Communication, AI Tools

## Projects

**HealthPay Incident Console** | *C#, .NET, SQL Server, Splunk, SSO, Docker*

- Built a **healthcare payment production-support console** that correlates transaction states, SQL records, application logs, and SSO events, guiding incident triage, root-cause analysis, remediation tracking, and stakeholder communication.

**Integration Reliability Lab** | *C#, ASP.NET Core, REST APIs, OAuth2, New Relic, Jenkins*

- Designed a **distributed integration testing and monitoring toolkit** that replays API failures, validates authentication flows, traces cross-service requests, detects recurring defects, and verifies automated fixes through CI pipelines.

**Chat2SQL** | *Java, TypeScript, SQL, PostgreSQL, MySQL, Docker* | Code

- Built an **AI-powered SQL development workspace** supporting 10+ databases, accelerating query authoring, schema exploration, and data troubleshooting through a unified Java and TypeScript interface with Docker-based deployment.